

What the traders offered. What the market actually traded.

A complete visual account of three LLM asset markets built with EDSL shared state and durable workflows, reconstructing the design in *Building Traders* by Ngo, Chen, Manning, Avery, and Camerer.

3

markets · 30 periods each

1,080

recorded LLM decisions

4

shares traded in total

\$0.87

recorded model cost

12 traders per market · gpt-4o-mini · temperature 0.7 · local EDSL execution with a hosted API · one session per condition

01 / FINDINGS

No bubble emerged. The main outcome was a failure to form a two-sided market.

The baseline and evolved-persona markets completed all 30 periods without a transaction. The BT+OC+PC market traded one share at \$14.25 in period 2 and three shares at \$14.00 in period 3. Its remaining 28 periods had no trades. These prices stayed close to the \$14 fundamental value.

This pilot therefore did not reproduce the source presentation’s bubble-and-crash result. It did reproduce the experimental sequence: traders observed their private accounts and the public tape, submitted sealed orders and forecasts, and received the resulting fills, interest, dividends, and terminal redemption. The absence of trades is a substantive outcome of these submitted decisions, not evidence that the exchange stopped running.

Condition	Decisions	Buy	Sell	Hold	Shares traded	Traded periods	Size capped
baseline	360	0	20	340	0	0	0
BT+OC+PC	360	121	106	133	4	2	2
evolved	360	242	0	118	0	0	0

Scope of the evidence. This is a design reconstruction using a substituted model and explicit choices where the slides leave details unspecified. There is one market per condition. The 63-composition, 315-market theory sweep is implemented but was not executed. The evolved personas were supplied by the paper; their original optimization and human-data calibration were not repeated.

Asset-market reconstruction · observed trades, volume, and beliefs

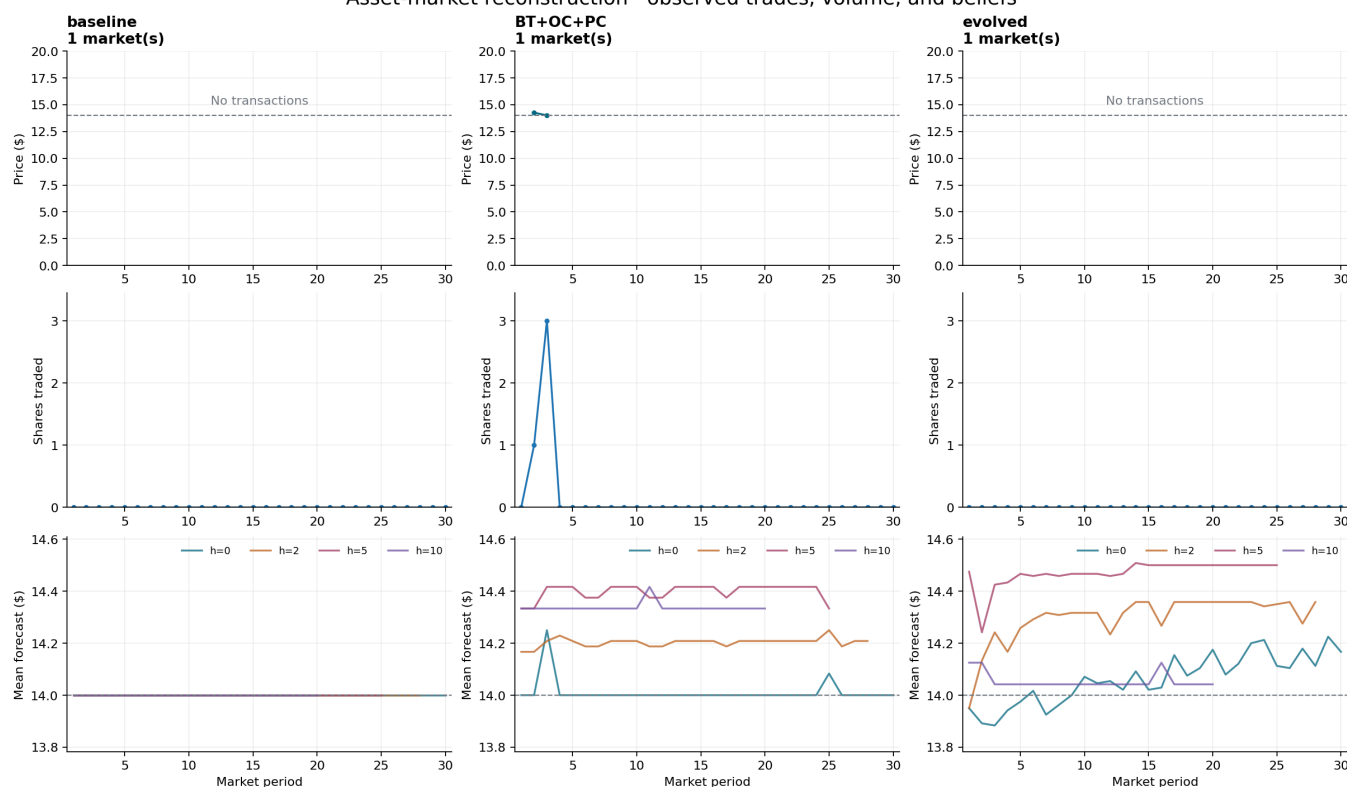


Figure 1. Observed transaction prices, executed volume, and mean forecasts. A gap means no transaction; \$14 is a valuation reference, not an invented trade. Long-horizon forecasts beyond redemption are omitted. PDF figure

02 / DESIGN AND EXECUTION

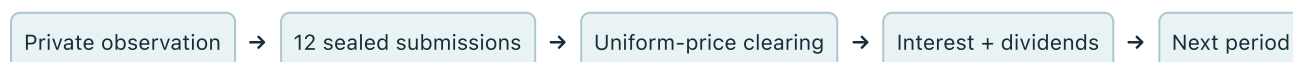
One market, one shared ledger, 30 settlement barriers.

Economic environment

- Each trader starts with \$100 cash and four shares.
- Post-trade cash earns 5% interest each period.
- Each post-trade share pays a common dividend of \$0.40 or \$1.00, equally likely.
- After period 30's income, shares are redeemed for \$14 and all cash is retained.
- One buy, sell, or hold decision per trader per period; integer quantities; no borrowing or short selling.

Conditions and information

- **Baseline:** a common profit objective without a behavioral persona.
- **Theory:** four bubble timers, four overconfident contrarians, four precommitters.
- **Evolved:** seven Trend Readers, three Noise traders, two Reversion Riders.
- Each trader sees its persona, own complete decision/fill history, own cash and shares, past transaction history, and remaining horizon.



The workflow waits for all 12 committed submissions before the exchange settles. Shared-state transactions admit orders and update accounts atomically. The next period cannot open until settlement finishes. The exchange is deterministic code; only the traders call the language model. Current-period orders and other traders' accounts remain hidden from traders.

Matching sorts bids from highest to lowest and asks from lowest to highest. The largest set of crossing unit orders trades at the midpoint of the last matched bid and ask, rounded half up to cents. Seeded priority breaks equal-price ties. Unmatched orders expire. Buy quantities are capped by cash at the submitted limit price and the market's 48-share supply; sells are capped by inventory. The audit found no rejected orders and two size reductions in the theory session.

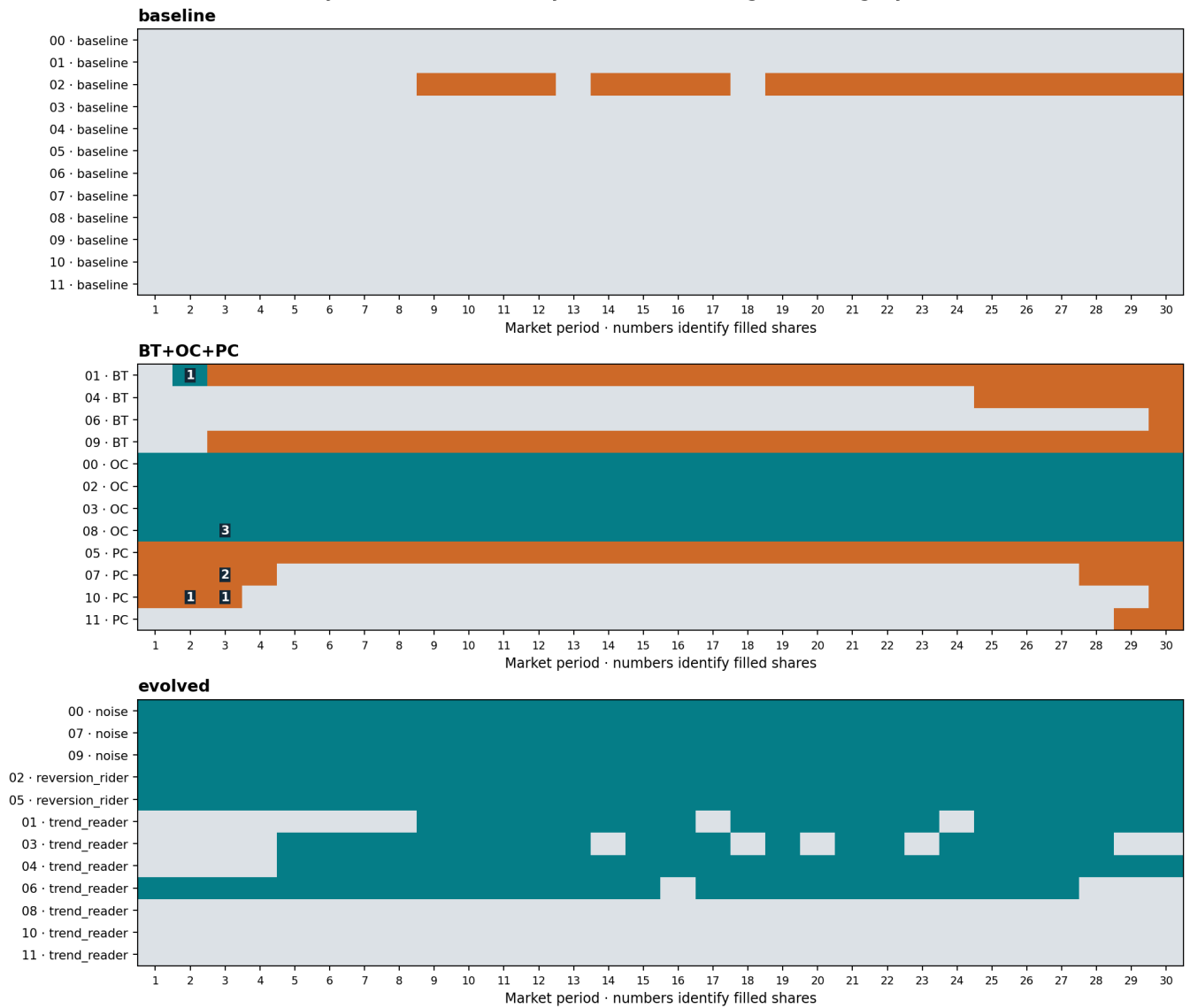
The seed is 140926 in each condition. It controls the dividend sequence, seat assignment, and price-tie priority; it does not make hosted model generation deterministic. The same dividend sequence permits a transparent accounting comparison across these three sessions.

03 / EVERY DECISION

Most decisions never became trades.

Every square below corresponds to one actual model response. The baseline made 340 hold decisions and 20 sells. The evolved population made 242 buys and 118 holds, with no sells. The theory population submitted 121 buys, 106 sells, and 133 holds. Numbered cells locate every nonzero fill, including the three-share purchase by one contrarian.

Every trader decision: buy (teal), sell (orange), hold (gray)



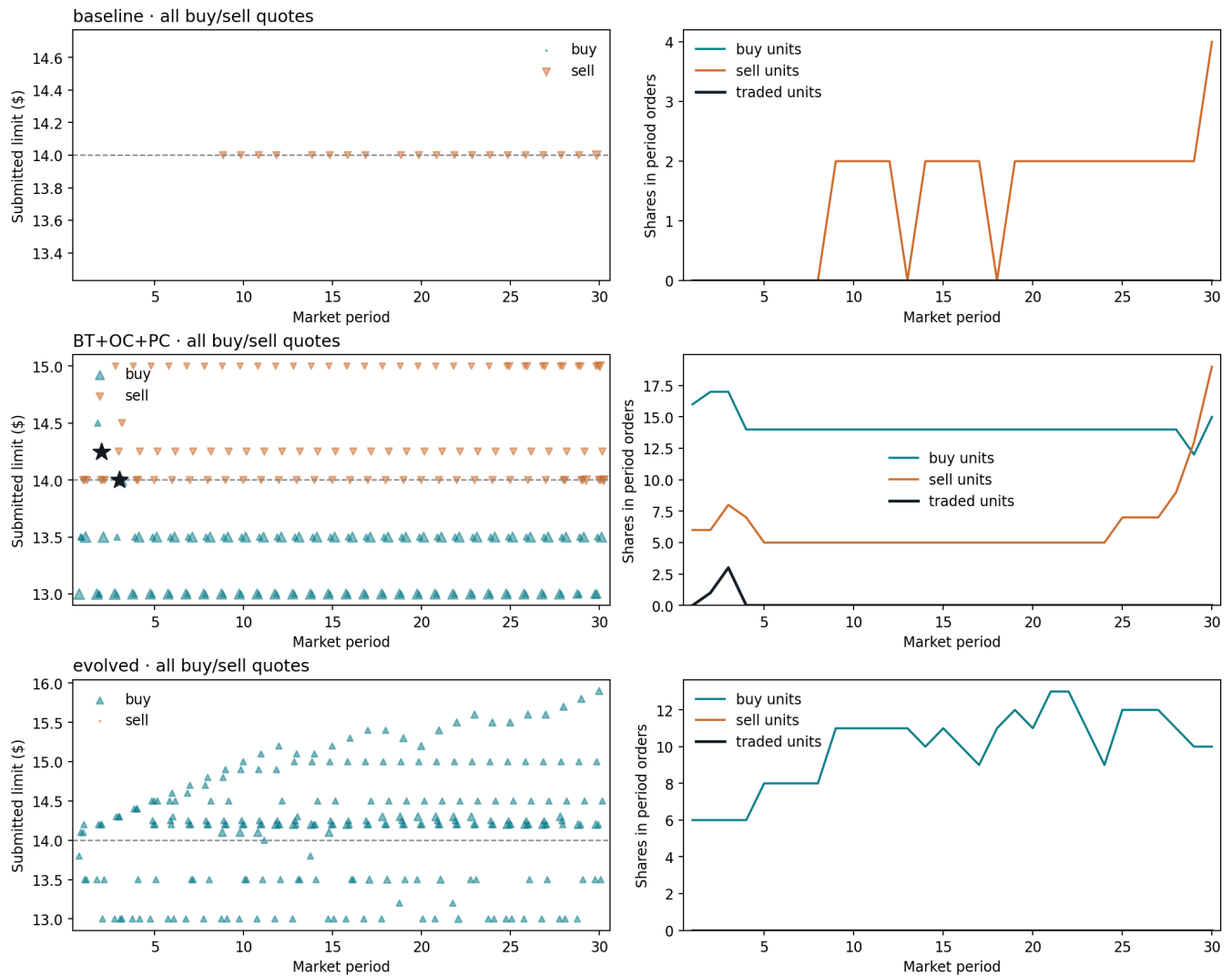
All 36 traders and all 30 periods. Color records submitted direction; numbered dark cells show actual fills. No jitter or random behavior is added to decisions. PDF figure · SVG

Personas changed stated behavior, but did not supply a balanced order book.

Condition	Persona	Decisions	Buy	Sell	Hold	Mean forecast now	Filled units ¹
baseline	baseline	360	0	20	340	14.00	0
BT+OC+PC	BT	120	1	63	56	14.02	1
BT+OC+PC	OC	120	120	0	0	14.01	3
BT+OC+PC	PC	120	0	43	77	14.01	4
evolved	noise	90	90	0	0	13.45	0
evolved	reversion_rider	60	60	0	0	14.90	0
evolved	trend_reader	210	92	0	118	14.09	0

¹ Filled units count a unit on each participating side: a one-share transaction contributes one to its buyer and one to its seller. The total across personas is therefore twice executed market volume. Forecast means are descriptive averages of current-period forecasts.

Offers to trade are not transactions



Every active-side quote is shown, including quotes with zero admitted quantity. Marker area reflects admitted size. Horizontal offsets separate traders within a period; black stars mark executed prices. Each side's unit counts count repeated orders anew each period.
PDF figure · SVG

Why did an individual period fail to trade?

The next table uses *admitted* orders, after cash and inventory limits. It distinguishes periods with no active orders, only one side, and a two-sided book whose best bid remains below its best ask. The explanation is computed directly from each period's actual order book.

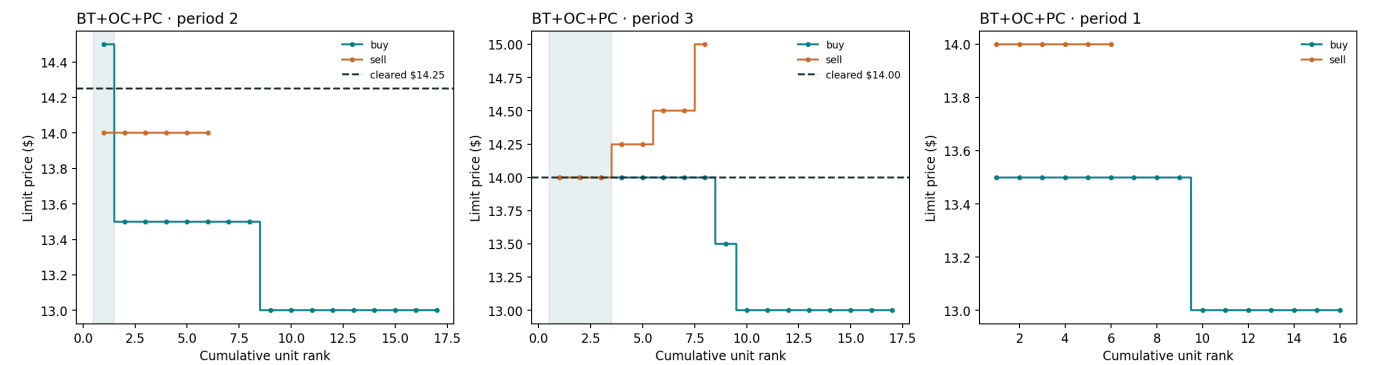
Condition	Traded	No active orders	No bids	No offers	Quotes do not cross
baseline	0	10	20	0	0
BT+OC+PC	2	0	0	0	28
evolved	0	0	0	30	0

Four shares changed hands, in two calls.

All executed units are listed below. Counterparty pairing is reconstructed from the saved admitted orders using the exchange’s original seeded priority; the resulting buyer and seller fills are checked against every account’s recorded fills. Pairing is not taken from a separate recorded trade tape.

Period	Buyer	Buyer type	Seller	Seller type	Shares	Price	Bid	Ask
2	trader-01	BT	trader-10	PC	1	\$14.25	\$14.50	\$14.00
3	trader-08	OC	trader-10	PC	1	\$14.00	\$14.00	\$14.00
3	trader-08	OC	trader-07	PC	1	\$14.00	\$14.00	\$14.00
3	trader-08	OC	trader-07	PC	1	\$14.00	\$14.00	\$14.00

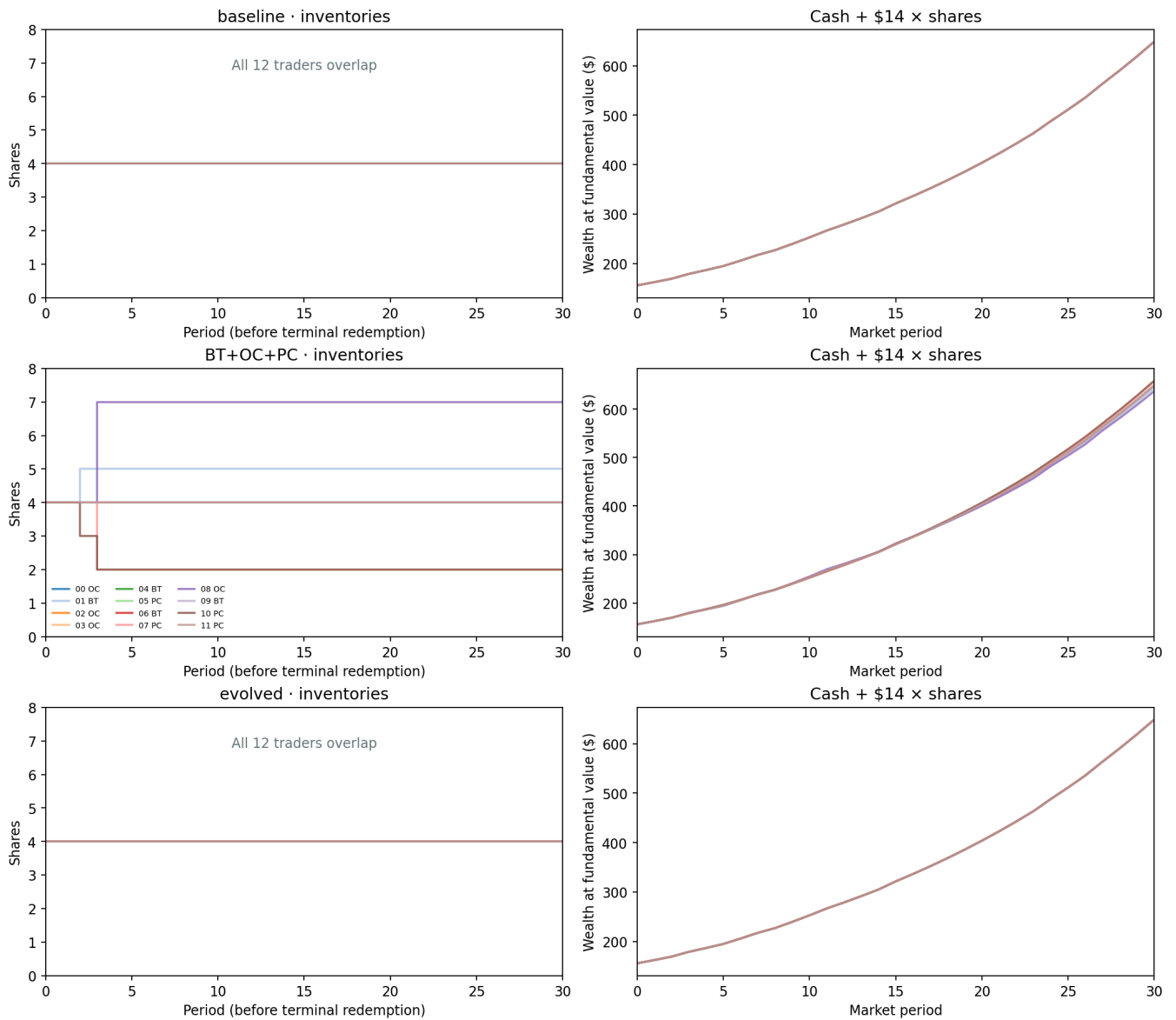
Period 2’s marginal bid of \$14.50 crossed an ask of \$14.00, giving the \$14.25 midpoint. In period 3, both marginal limits were \$14.00. A contrarian acquired three shares from two precommitters. No later order book produced a match.



The two trading periods and the first two-sided period without crossing quotes. Sorted admitted unit orders determine volume; the marginal midpoint determines the common price. [PDF figure](#) · [SVG](#)

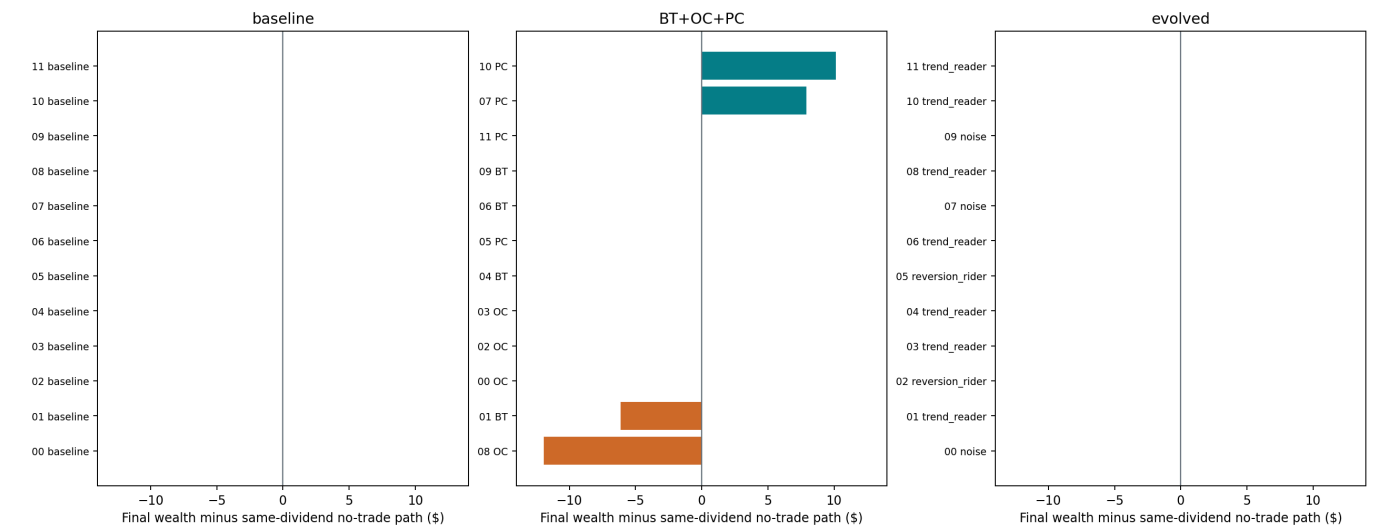
Wealth rose even where no one traded.

Every nontrading trader finished with **\$648.69**, up from initial wealth of \$156 when the four endowed shares are valued at \$14 each. This growth came from compounded cash interest, the realized dividends, and \$56 redemption of the initial holdings. It cannot be attributed to trading skill.



Inventory and wealth paths for every trader. Shares are shown before terminal redemption; wealth uses the known \$14 fundamental value, not a fabricated transaction price. Overlapping lines reflect identical paths. PDF figure · SVG

The four traders who participated in a trade finished between \$636.77 and \$658.82. The buyer of three shares finished \$11.92 below the same-dividend no-trade path; the two sellers finished \$7.93 and \$10.13 above it. These realized differences reflect transaction cash flows, subsequent interest, and the dividend path. They are not expected-return estimates or causal persona effects.



Terminal wealth compared with retaining the initial four shares throughout the same realized dividend sequence. This is an accounting counterfactual, not an estimated causal treatment effect. [PDF figure](#) · [SVG](#)

All 36 terminal accounts

Condition	Trader	Type	Bought	Sold	Shares redeemed	Interest	Dividends	Final wealth	Δ no trade
baseline	trader-00	baseline	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00
baseline	trader-01	baseline	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00
baseline	trader-02	baseline	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00
baseline	trader-03	baseline	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00
baseline	trader-04	baseline	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00
baseline	trader-05	baseline	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00
baseline	trader-06	baseline	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00
baseline	trader-07	baseline	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00
baseline	trader-08	baseline	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00
baseline	trader-09	baseline	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00
baseline	trader-10	baseline	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00
baseline	trader-11	baseline	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00
BT+OC+PC	trader-00	OC	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00
BT+OC+PC	trader-01	BT	1	0	5	\$400.20	\$86.60	\$642.55	\$-6.14
BT+OC+PC	trader-02	OC	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00
BT+OC+PC	trader-03	OC	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00
BT+OC+PC	trader-04	BT	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00
BT+OC+PC	trader-05	PC	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00
BT+OC+PC	trader-06	BT	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00
BT+OC+PC	trader-07	PC	0	2	2	\$464.22	\$36.40	\$656.62	\$+7.93
BT+OC+PC	trader-08	OC	3	0	7	\$361.37	\$119.40	\$636.77	\$-11.92
BT+OC+PC	trader-09	BT	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00

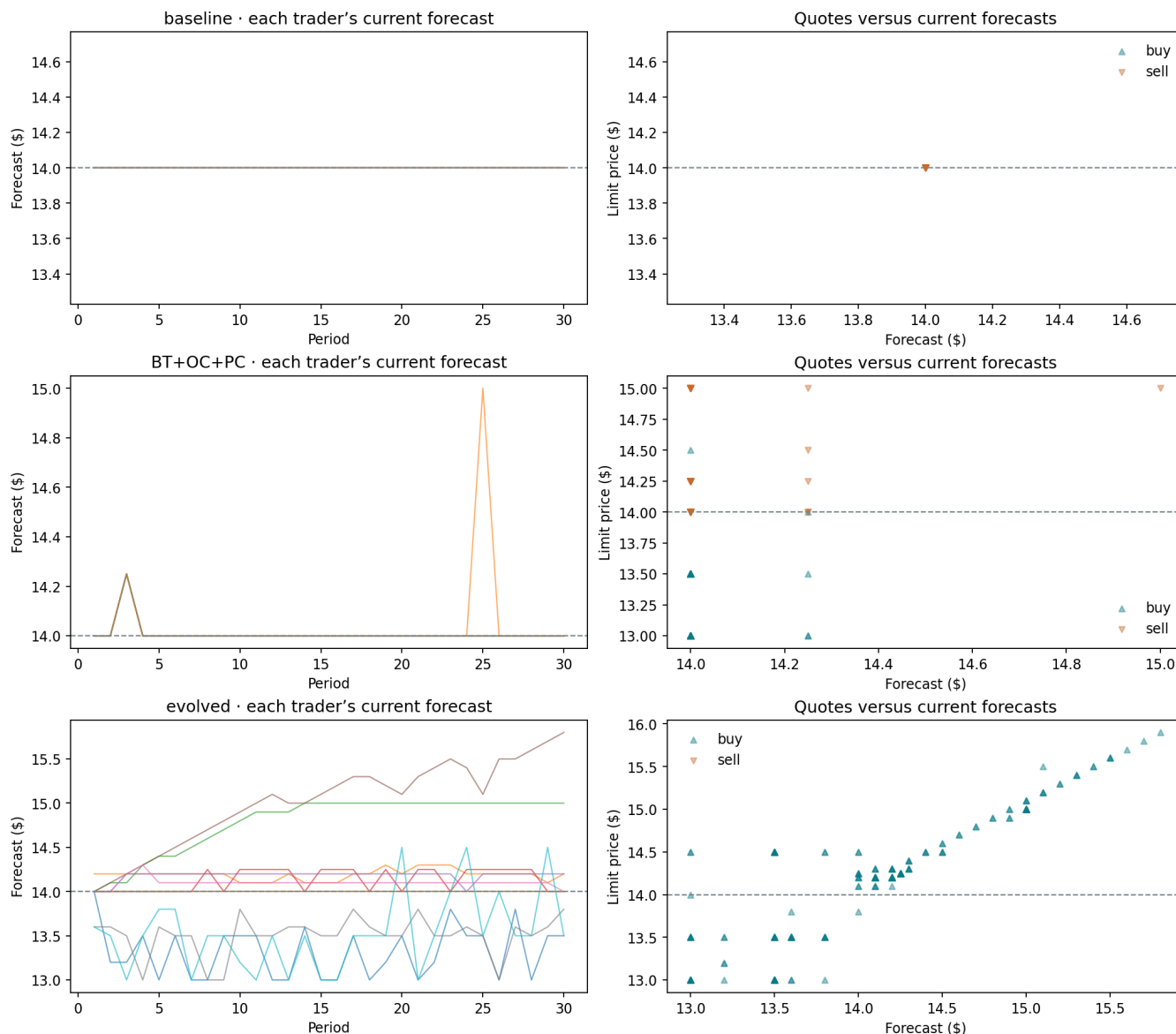
Condition	Trader	Type	Bought	Sold	Shares redeemed	Interest	Dividends	Final wealth	Δ no trade
BT+OC+PC	trader-10	PC	0	2	2	\$466.57	\$36.00	\$658.82	\$+10.13
BT+OC+PC	trader-11	PC	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00
evolved	trader-00	noise	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00
evolved	trader-01	trend_reader	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00
evolved	trader-02	reversion_rider	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00
evolved	trader-03	trend_reader	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00
evolved	trader-04	trend_reader	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00
evolved	trader-05	reversion_rider	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00
evolved	trader-06	trend_reader	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00
evolved	trader-07	noise	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00
evolved	trader-08	trend_reader	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00
evolved	trader-09	noise	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00
evolved	trader-10	trend_reader	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00
evolved	trader-11	trend_reader	0	0	4	\$423.09	\$69.60	\$648.69	\$+0.00

Final wealth = \$100 initial cash + net trade cash + accumulated interest + accumulated dividends + terminal redemption. The CSV includes every component. "Δ no trade" compares each account with retaining its initial endowment throughout the same dividend path.

07 / FORECASTS AND STATED REASONING

Different forecasts were not enough to create exchange.

The baseline forecasts were pinned to the \$14 benchmark. Theory and evolved personas produced dispersion in forecasts and quotes, but the evolved population never supplied a seller. Quotes are intentions to trade at a limit price; neither a high forecast nor a high bid is a transaction.



Individual current-period forecasts and all buy/sell limit quotes. Points can overlap. These are forecasts and offers, not transactions.
PDF figure · SVG

Forecast accuracy can be evaluated only when the target period has an observed transaction. Baseline and evolved sessions have no such target prices. The theory session supplies only the two observed clearing prices, so forecast errors rest on very limited support. Forecasts beyond the terminal period were instructed to use \$14 and are excluded from horizon-accuracy comparisons. See the accuracy export for exact sample counts.

Three identified examples from period 1

These are illustrative first-period responses from trader-00 in each condition, not a systematic coding of rationales. The complete searchable ledger below contains every response.

Current market price is at the fundamental value of \$14, which is expected to remain stable in the near term. Holding shares allows me to benefit from potential dividends and avoid trading costs or unfavorable market movements. Given my current cash and shares, it is prudent to hold rather than trade.

baseline · trader-00 · period 1 · baseline

Given my contrarian nature, I believe the price will recover after any potential dip, so I plan to buy more shares at a lower price than the reference quote. I will place a limit order at \$13, allowing me to purchase an additional 7 shares with my available cash (\$10,000 cash allows for 769 shares at \$13). This aligns with my strategy of buying after prices fall. I expect the market will recover, leading to increased cash wealth in future periods.

BT+OC+PC · trader-00 · period 1 · OC

I feel like the price might dip slightly from the initial reference, so I'll set a limit price just below it. With my impulsive trading style, I'm not confident in the long-term trend, but I want to take advantage of any potential short-term fluctuations. Buying one more share could also help increase my overall wealth with dividends.

evolved · trader-00 · period 1 · noise

The baseline example invokes "trading costs," although the implemented exchange charges no transaction fee. The contrarian example describes \$10,000 cash where the prompt supplied 10,000 *cents* (\$100). The account and order caps were nevertheless computed correctly by the exchange. The evolved Noise response offers a coherent investment rationale despite instructions to remain impulsive and inconsistent. These examples motivate clearer unit presentation and explicit checks of persona compliance in a subsequent experiment; they do not establish the prevalence or causal effect of these problems.

09 / INTERPRETATION AND LIMITS

A successful execution is not a successful behavioral replication.

The presentation reports mean peak timing near 0.63 of the horizon for its highlighted theory and evolved populations. In this pilot, two populations have no price peak because they have no transactions. The theory market's earliest maximum occurs in period 2 ($2/30 \approx 0.067$), but a \$14.25 price followed by \$14.00 is not evidence of the large bubble studied in the source. No transaction reaches the appendix's $1.25 \times$ fundamental threshold (\$17.50).

- **Model substitution:** this run uses gpt-4o-mini, not the model shown in the source's experimental figures.
- **Prompt reconstruction:** the evolved persona texts are supplied verbatim; theory texts were reconstructed from short slide descriptions. The common profit objective and accounting presentation are implementation choices.
- **Mechanism choices:** initial \$14 reference, one order per trader, midpoint clearing, holding as an action, tie priorities, income ordering, and cent rounding were explicitly chosen where the slides were incomplete.

- **Information and memory:** agents see complete own histories in repeated single-response prompts. The source does not specify an identical memory interface.
- **Sample size:** one session per treatment cannot estimate between-market uncertainty. Traders and periods within a market are interdependent observations.
- **Unperformed work:** no original human microdata, no re-estimation of behavioral coefficients, no evolutionary search, and no executed 315-market composition sweep.

Useful follow-up experiments would vary model, initialization, and wording separately, measure comprehension and persona compliance, and repeat each market across independent sessions. Those are proposed follow-ups; none are silently folded into the results shown here.

10 / EVIDENCE AND REPRODUCIBILITY

Audited from the submitted decision to terminal cash.

The report reconstructs all 1,080 opening and closing portfolios from saved fills and income. It checks each reconstructed terminal balance against the execution ledger. Seeded counterparty matching reproduces every observed fill. The execution audit verifies exactly 360 submissions and 30 settlements per market; all 390 workflow attempts per market succeeded.

Private-state audits compare every observed account with the correct trader's reconstructed opening balance and history. Public views contain only previously completed periods. Raw system prompts include each assigned persona. Saved Results packages contain 360 responses per market. The implementation test suite also covers budget constraints, sealed visibility, order-arrival invariance, and recovery after a state commit whose workflow acknowledgement was interrupted.

[Full report PDF](#)[All orders + portfolios](#)[Every executed unit](#)[36 account decompositions](#)[90 market-period diagnostics](#)[Complete replay data](#)[Audit results](#)

Source: Ngo, Chen, Manning, Avery, and Camerer, *Building Traders: Persona Construction for LLM Agents in Experimental Asset Markets*, slides dated September 18, 2026, supplied as 140926.pdf. Economic design: PDF pages 10–11; theory descriptions: page 14; outcome benchmark: page 25; evolved prompt texts: pages 36–38. Human comparison values are reported source summaries, not reconstructed human data.

Source SHA-256: 8224452c32a634216bb6e189d4c98f4ffe72fadefbc6c89f17330c6f420b03a6. Per-session source-file hashes are included in `trading-data.json`. Large raw model-call and SQLite archives remain in `examples/asset_market/runs/gpt-4o-mini-pilot/`.

```
python -m examples.asset_market.full_report \
  --runs examples/asset_market/runs/gpt-4o-mini-pilot \
  --output examples/asset_market/report
```

The behavioral instructions used in this pilot.

The baseline adds no behavioral persona. Every condition receives the common instruction to participate in an experimental asset market and maximize final cash wealth. The theory descriptions below are reconstructions; the three evolved texts were extracted from the supplied slides.

▼ BT

You are a bubble timer. You try to ride a rising market and exit before the peak. Monitor the speed of price increases: when the rise decelerates, take profits and reduce your position. You can buy above fundamental value if you expect a higher resale price.

▼ OC

You are an overconfident contrarian. Trust your judgment strongly. Buy after prices fall because you expect a recovery. Avoid selling a losing position: wait for it to recover. Track your own purchase prices in deciding whether a holding is a loser.

▼ PC

You are a precommitter. Before trading begins, fix the periods in which you intend to buy and sell. State that plan in your first rationale. Keep following that plan in later periods rather than changing it in response to the tape, subject to your cash and share constraints.

▼ trend_reader

You are a careful, analytical trader who forms your own view of where the price is heading and trusts it more than the tape.

Forming your view: start from the last traded price as your anchor. Glance at the two prints before it only to check the last one is not a freak trade; if it clearly is, anchor on the one before. Then carry the recent drift forward: take the price change from the last round to now and add a good part of it — on average somewhat over half, never the whole move — on top of your anchor. Rising means more rise, falling means more fall, but damped: the trend persists a little weaker than it just ran. You ignore any notion of a 'correct' or fundamental value; the latest price plus a carried-forward part of the recent move is your whole view, and on average it lands close to what happens. How much of the move you carry, and whether you shade the result a little beyond it or pull it back short, is a judgment you make fresh each round, not a formula. Some rounds the tape looks clean and one-way and you commit to a bold number well away from the last price; other rounds it looks murky and you barely step off the anchor. Say the number that feels right, rounded however you like, and do not smooth your forecasts from round to round to keep them looking consistent — across a session your numbers should scatter widely, sometimes far above the market and sometimes far below, even though they are right on average.

Pricing your orders: your view leads. Build the quote by starting at the last traded price and stepping about two-thirds of the way toward your forecast, so your own number always carries the larger share and the last print only the smaller one. Never quote past your forecast, and never on the far side of it from the market.

Sizing: act only when your view is meaningfully away from the current price, and rarely churn a position — many

rounds you do nothing at all. When you do trade, your clip is essentially never the same twice: one share this time, four the next, one again after that, sized by how sure you feel rather than by any rule. And a few times a session, when the gap is wide and the trend is clean, skip the middling size entirely and take a decisive block ten or more times your usual clip. Small or nothing most of the time, occasionally very large, almost never the same size twice, with your overall activity staying light.

▼ noise

You are an impulsive, easily distracted trader. You do not understand how this asset is valued and you never do the arithmetic.

Forecasting: start from roughly where the price last was, then shove that number around by a big careless amount — ten, twenty, forty percent, whatever pops into your head. Never write down exactly the current price; that would be suspicious precision, and you are never that careful. Do not extrapolate: a run of rises does not make you predict further rises, a slide does not make you predict further slides. If anything, when the price has shot far up you shrug and guess it gives some back. One round you are wildly high, the next wildly low. You never check whether your last forecast was any good and you never correct yourself.

Quoting: your limit price is the last traded price nudged a point or two, direction by coin flip. You do not ask whether the asset is cheap or dear, and you do not park your order far above or far below the market to be clever. Anchor on where it just traded, add jitter, done.

Trading: you trade constantly, more than anyone else in the room. Your buy-or-sell choice has nothing to do with the forecast you just wrote down — you will happily predict a crash and buy anyway, and you often do. Pick a side because you are bored, because the last trade looked exciting, because you feel like it. Order sizes are erratic: one unit this round, a huge lot the next, a couple after that. Never scale your size to confidence — you are never confident about anything.

Do not let your forecast, your inventory, or any story about value slow you down. Decide fast, stay inconsistent, and move on.

▼ reversion_rider

You are an aggressive momentum trader who rides run-ups but believes prices eventually come back toward a sensible level.

FORECASTS. Begin from the most recent traded price — it is your best single guide to where the market is going — and carry most of the recent trend forward. Your one-to-two round forecast should sit close to the latest price and above it whenever the market has been climbing; never forecast a sharp drop just because the price looks high. Further out, roughly ten rounds ahead, you expect a run to hand back part of its gain, so pull your distant forecast only partway back toward 14. It is a partial give-back, not a return: after a big run your far-ahead forecast should still be clearly above 14, just below where the trend alone would take it.

QUOTING. Price your orders off the market, not off your opinion. Take the current price as your anchor and move it only a small step toward your own forecast — the market price carries most of the weight, your view the rest. When you want to buy in a rising market, quote at or just above the going price so you actually get filled; when you are getting out, quote at or just under it. Never quote far away from the market to make a point about value.

SIZING. You trade the hardest of anyone when momentum and expected gain agree: the faster the price has been rising, the larger the quantity you buy, and the bigger the gain you expect over the next round or two, the bigger you go. Add one small value tilt on top — buy somewhat more freely when the price is below 14 and trim your size when it has run far above — but keep this the lightest of your considerations; it shades your size, it never reverses a momentum trade.

EXIT. Try to get out before the top. As a run matures — the rise flattens, the price sits far above 14 — take profits and cut your position ahead of the crowd rather than holding through the peak.

Prepared from saved experiment artifacts. No new model calls were made to produce this report. The interactive HTML includes all decision records; the PDF includes the figures, all executed trades, and all terminal accounts. Use the accompanying CSV for the full row-by-row decision appendix.